

ASILAH MARYAM

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I am a mechanical engineer interested in biomimicry, automation, and human-centered design. From Mars rovers to surgical instruments, I aim to create systems that serve the people and environments they're built for.

EDUCATION

University of California, Berkeley

May 2027

M.Eng. Mechanical Engineering, Product Design Specialization

Dean's List Award

University of California, Los Angeles

June 2026

B.S. Mechanical Engineering, Minor in Digital Humanities

GPA: 3.89 | Cum Laude, Dean's Honor, Academic Excellence Scholarship

Relevant coursework: CAD, MATLAB, Statics & Strength of Materials, Dynamics, Finite Element Analysis, Compliant Mechanisms Thermodynamics, Fluid Dynamics, Control Systems

SKILLS

Computer-Aided Design (CAD), SolidWorks, Engineering Drawings, Finite Element Analysis (FEA), Rapid Prototyping, 3D Printing, Laser Cutting, Machining/Manufacturing, DFM, DFA, GD&T, Automation, Programming, Arduino (C++), Java, MATLAB, Python (data analysis), Microcontrollers, Leadership, Project Management

PROFESSIONAL EXPERIENCE

Intuitive Surgical, Sunnyvale, CA

June 2025 - September 2025

R&D Mechanical Engineering Intern (Multiport Endoscope Team)

- Evaluated methods of pressure equalization by designing and manufacturing an autoclavable DUT using GD&T principles to improve endoscope lifetime
- Identified sources of variation in button actuation via worst-case and RSS tolerance stackup analysis
- Developed a test plan and performed data analysis, delivering design suggestions adopted by the team

Amazon Lab 126, Sunnyvale, CA

June 2024 - September 2024

Hardware Dev Engr Intern (Prototyping, Automation, and Calibration Team)

- Engineered a suspended automation system using a 4-cable driven parallel robot (CDPR) configuration for flexible Amazon device testing
- Performed mechanical system analysis in MATLAB to ensure structural integrity and optimize performance
- Programmed motion algorithms via Arduino IDE, applying geometric functions to improve payload path efficiency
- Sourced servos, pulleys, and belts from 3 suppliers based on mechanical analysis specifications

Amazon Lab 126, Sunnyvale, CA

June 2023 - September 2023

AFE Hardware Intern (Prototyping, Automation, and Calibration Team)

- Designed a cost-effective (\$250) automated button-pushing system for Fire TV remote testing, cutting remote rework time from 2 weeks to hours
- Researched and evaluated 4 actuators for optimal cost, durability, and compatibility
- Rapid-prototyped a button pusher mechanism in SolidWorks for compatibility with existing remotes

LEADERSHIP

urc@ucla, Los Angeles, CA

February 2024 - June 2026

Co-founder and Education Lead

- Co-lead an all-women's robotics team of 40+ members across design, manufacturing, programming, electrical, and marketing sub-teams, building on a robotics team first founded in high school
- Develop hands-on workshops preparing members for the 2025 and 2026 University Rover Challenges

PROJECTS

Calypso and Pandora's Box – urc@ucla

September 2025 - June 2026

- Developed the electrical system for a Mars rover and wired and programmed motors, actuators, sensors, and long-range communications; designed differential mechanism in SolidWorks
- Manufactured arm, suspension, and chassis components and worked with the team for final assembly

Jetsam – Stanford TreeHacks

February 2026

- Built, wired, and programmed a trash-collecting water robot from reclaimed materials with integrated object detection